

Project Design Phase

Problem – Solution Fit

Parameter	Details
Date	13 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	2 Marks

Problem – Solution Fit Overview

The Problem–Solution Fit validates that a real customer problem exists and that the proposed solution effectively addresses it. For this project, the fit is evaluated across fruit vendors, farmers, distributors, and food quality inspectors who face challenges in identifying and separating rotten fruits efficiently.

Target Customers

- Fruit vendors and shop owners
- Farmers and agricultural producers
- Fruit distributors and warehouse operators
- Food quality inspectors and supply chain managers

Customer Problems (Key Pains)

- Difficulty identifying rotten fruits among large batches
- Manual sorting is time-consuming and labor intensive
- Risk of selling spoiled fruits to customers
- Financial losses due to fruit spoilage and wastage
- Lack of automated systems for quick fruit quality detection

Proposed Solution

A **Machine Learning–based Smart Sorting System** that provides:

- Automated fruit classification using image recognition
- Detection of fresh and rotten fruits using deep learning models
- Fast and accurate prediction through a simple web interface
- Visual output indicating fruit quality status
- Improved sorting efficiency and reduced manual effort

Problem – Solution Fit Mapping

Customer Problem	Solution Feature	Expected Impact
Difficulty identifying rotten fruits	Image-based fruit classification using ML	Faster and more accurate fruit sorting
Manual inspection process	Automated detection system	Reduced labor and time consumption
Mixed quality fruits in batches	Real-time fruit quality prediction	Improved product quality control
Financial losses due to spoilage	Early detection of rotten fruits	Reduced waste and increased profitability

Channels & Adoption

- Deployment as a **web-based application for fruit vendors**
- Integration with **automated fruit sorting machines (future scope)**
- Use in **agricultural markets and warehouses**
- Adoption by **food quality monitoring systems**

Success Metrics

- Fruit classification accuracy
- Reduction in manual sorting time
- Decrease in fruit wastage
- System prediction speed (<1 second per image)
- User satisfaction among vendors and distributors

Future Improvements

- Real-time fruit detection using live camera input
- Integration with automated conveyor belt sorting machines
- Deployment as a mobile application for farmers
- Expansion to detect multiple fruit diseases and quality defects
- Cloud deployment for large-scale agricultural monitoring